Detecting a Faint Signal
Exercise Guide for 5A

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Mathematical Methods for Physicists II

Python Notebook Guide

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This guide follows Notebook 5A from its coin warm-up through the pulse fit and the noise-only location search.

1 Start with the coin warm-up

The first two sections update a Beta(2,2) prior after eight heads in twelve flips. One cell computes the Beta(10,6) posterior directly. The next evaluates the likelihood and prior on a grid; the two curves overlap. Check that the predicted probability of heads on the next flip is 0.625. The pulse exercise uses this same grid method with two unknowns rather than one.

2 Model the pulse and its noise

The signal is $Aq_i(c)$, where $q_i(c) = \exp[-(x_i - c)^2/(2w^2)]$. The template peaks at one, so A sets height, c sets location, and the known $w = 0.50$ sets width. The 80 positions span $[-5, 5]$. Independent Gaussian noise has known $\sigma = 1$. The simulation uses $A = 1.5$ and $c = 0.7$; the fit does not use those values. Under the noise-only model the mean is zero.

A single point may reveal little. A template combines nearby points, giving more weight where the pulse should be strongest. Expanding the squared residuals gives

$$\log \frac{L(A, c)}{L_0} = \frac{A \sum_i y_i q_i(c) - \frac{1}{2} A^2 \sum_i q_i(c)^2}{\sigma^2}.$$

The first sum measures alignment; the second penalizes too large a prediction. The code computes both for each candidate centre, then uses broadcasting to combine them with every amplitude.

3 Read the parameter plots

Uniform priors span $A \in [0, 4]$ and $c \in [-2, 2]$. Rows represent amplitude, columns represent centre. Summing over columns gives amplitude mass; summing over rows gives centre mass. The heat map shows plausible pairs and the marginal curves show each parameter's uncertainty after averaging over the other. The representative curve uses the two posterior means and does not show a predictive band.

For the supplied seed, mean amplitude is 1.397 with central 95% interval [0.624, 2.153]; mean centre is 0.780 with interval [0.312, 1.257]. All these summaries assume a signal. A continuous amplitude prior assigns zero mass to the single point $A = 0$ regardless of the observations, so the amplitude interval is not a test of the noise-only model.

4 Compare models and check the prior

The code averages the likelihood ratio over the signal prior with equal-area midpoint cells and log-sum-exp. It gives $B_{10} \simeq 55.44$, or about 0.982 signal-model probability with equal prior odds. With wider bounds $A \in [0, 8]$, $c \in [-4, 4]$, the same observations give $B_{10} \simeq 13.89$. The wider prior assigns more mass to pulses that fit poorly. Doubling the number of cells along each axis preserves the original resolution.

5 Calibrate the full search

Under noise alone, a projection $\sum_i y_i q_i(c)$ at a fixed centre has standard deviation $\sigma\sqrt{\sum_i q_i(c)^2}$. Divide by this scale to compare templates. The code clips negative matches at zero because it searches for positive pulses, then retains the largest match over centres.

A large match is more common after scanning many locations than at a location chosen in advance. Overlapping templates give correlated matches. The simulation accounts for both effects by running the same search on 400 noise-only datasets. The histogram contains one maximum per dataset. Its vertical line marks the observed maximum.

If r simulated maxima reach or exceed the observed maximum, the rank-based value is $(r + 1)/401$. Here it is $2/401 \simeq 0.005$. The minimum reportable value is $1/401 \simeq 0.0025$, so the experiment has limited tail precision. This statistic is calibrated under noise alone, while the Bayes factor averages across the signal prior.

6 Try it yourself

1. Set the generating amplitude to zero. Compare the fitted pulse, which assumes a pulse exists, with the evidence for that model.
2. Change the amplitude and centre bounds one at a time while keeping cell spacing fixed.
3. Repeat with multiple noise draws before drawing conclusions about detection performance.

For real observations, check the baseline, width, noise level, independence, and how the search range was chosen.